Introduction To Industry 4.0 And Industrial Internet Of Things

Assignment-9

Number of questions: 15 **Total marks: 15 X 1= 15**

1. Which of the following is not a valid reason for employing fog computing in IIoT?

a. Consistent release of data from sensors and machines
b. Need for immediate action and quick response
c. Devaluing existing functionalities of IIoT
d. Data may be critical as well as time-sensitive

Answer: (c)

Justification: Please refer to Slide 2 of Lecture 1 of Week 9

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2. Which of the following statements is false about the fog-enabled IIoT for Smart Grid and Power
Industry?

a. Dynamic demand of appliances
b. Unidirectional communication between the consumer and supplier
c. Power supply is provided from micro-grids, local distribution companies
d. Advanced metering infrastructure for bi-directional communication

Answer: (b)

Justification: Please refer to Slide 10 of Lecture 1 of Week 9

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3. Nebbiolo Technologies, who is one of the pioneers of Fog Computing, brought together IT and OT for real-time services and rendered optimized solutions with smarter decision making capability in lower cost. Which of the following products is not produced by Nebbiolo Technologies?

a. fogOS
b. fogNode
c. fogSM
d. fogBCD

Answer: (d)

Justification: Please refer to Slide 20 of Lecture 1 of Week 9

4. SONM is a Decentralized Fog Computing Platform that renders distributed cloud services with fog as the backend. Which of the following facilities is not supported by SONM?

a. Blockchain infrastructure
b. Video streaming
c. Machine learning
d. Digital triplet

Answer: (d)

Justification: Please refer to Slide 22 of Lecture 1 of Week 9

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5. “In R Programming, it is compulsory to specify the datatype at the time of variable declaration.” State true or false.

a. True
b. False

Answer: (b)

Justification: Please refer to Slide 6 of Lecture 2 of Week 9

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6. Which of the following can be used to calculate a^b in R programming?

a. a+b
b. a/b
c. a^b
d. a@b

Answer: (c)

Justification: Please refer to Slide 7 of Lecture 2 of Week 9

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7. Which of the following operands is used to create series of numbers for a vector in R programming?

a. <=
b. #

c. :
d. !

Answer: (c)

Justification: Please refer to Slide 11 of Lecture 2 of Week 9

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8. Which of the following is/are supported by the Julia programming language?

a. Tuples
b. Dictionary
c. Arrays
d. All of the above

Answer: (d)

Justification: Please refer to Slide 21-23 of Lecture 2 of Week 9

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9. Which of the following represents the essential characteristics of cloud computing according to NIST?

a. On-demand self service
b. Resource pooling
c. Broad network access
d. All of the above

Answer: (d)

Justification: Please refer to Slide 3 of Lecture 3 of Week 9

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10. Why is it necessary to perform integration which is a part of data pre-processing in the context of Data management?

a. Mitigating redundancy
b. Combining data from various sources
c. Cleaning incomplete data
d. Identifying inaccurate data

Answer: (b)

Justification: Please refer to Slide 10 of Lecture 3 of Week 9

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11. “There are three main NoSQL databases. They are Key-value databases, column-oriented databases, and document-oriented databases.” State true or false.

    a. True
    b. False

Answer: (a)

Justification: Please refer to Slide 11 of Lecture 3 of Week 9

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12. “In Hadoop Master/Slave Architecture, the master serves read and write requests from the file system’s clients whereas slave executes operations like opening, closing, and renaming files and directories.” State true or false.

    a. True
    b. False

Answer: (b)

Justification: Please refer to Slide 22 of Lecture 3 of Week 9

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13. Which of the following is false about the BCube topology of the data center network.

    a. Follow switch-oriented practice, rather than the server-centric approach
    b. Provides multiple parallel short paths between any pair of server
    c. Places intelligence on modular data center (MDC) servers
    d. Forms multiple edge-disjoint server spanning tree

Answer: (a)

Justification: Please refer to Slide 11 of Lecture 4 of Week 9

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14. Which of the following protocols is used by the SDN?

    a. OpenByte
    b. OpenFlow
    c. WideFlow
    d. ClosedByte

Answer: (b)

Justification: Please refer to Slide 4 of Lecture 5 of Week 9

15. Which of the following is incorrect about Software-defined networks (SDN)?

a. SDN increases complexity of traditional networks by separating the control and data planes
b. SDN and NFV are separate paradigms
c. SDN provides a centralized view of the whole network which improves controllability, efficiency and network stability
d. SDN can provide seamless transition of data routes, self-organization, configuration optimization and smart transmissions

Answer: (a)

Justification: Justification: Please refer to Page 316 of the book Introduction to IoT by S Misra, A Mukherjee, A Roy by Cambridge University Press